

PROJECT REPORT

Islamabad Rehab Center (IRC)

Universal Dynamic Web Portal & RBAC System

Submitted by

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Islamabad Rehab Center (IRC)

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I, **Haseeb Javed**, hereby declare that the project report entitled "**Islamabad Rehab Center (IRC) - Universal Dynamic Web Portal & RBAC System**" submitted to the Department of Business Administration, University of Sahiwal, is my original work. This project has not been submitted previously for the award of any other degree, diploma, or fellowship at this or any other university. All sources of information and assistance received during the course of this work have been duly acknowledged.

Haseeb Javed

DECLARATION

This is to declare that the project entitled "**Islamabad Rehab Center (IRC) - Universal Dynamic Web Portal & RBAC System**" is an original work done by the undersigned, in partial fulfillment of the requirements for the degree "**Bachelors of Business and Information Systems (BBIS)**" at the Department of Business Administration, University of Sahiwal, Sahiwal.

All the analysis, design, and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

Haseeb Javed

ACKNOWLEDGEMENTS

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Lastly, I extend my heartfelt thanks to my parents, family, and friends for their endless patience, moral support, and prayers, which kept me motivated during the challenging phases of this project.

DEDICATION

"We would like to dedicate this accomplishment to our beloved and caring Parents, friends, and to our teachers with the support of whom I am standing at this step of my life stairs."

"We also dedicate this dissertation to Mr. Umair Waqas who has supported us throughout the process."

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LIST OF ABBREVIATIONS

Abbreviation	Full Meaning / Definition
IRC	Islamabad Rehab Center
RBAC	Role-Based Access Control
PDO	PHP Data Objects
SQL	Structured Query Language
CRUD	Create, Read, Update, Delete
CNIC	Computerized National Identity Card
XSS	Cross-Site Scripting
PDF	Portable Document Format
UI / UX	User Interface / User Experience

ABSTRACT

Rehabilitation facilities handle critical, highly sensitive clinical data, including patient medical profiles, substance detox stages, and therapy progress notes. Managing these variables with manual logs or fragmented systems leads to security risks, communication delays, and audit inconsistencies. This project presents a **Universal Web Portal & RBAC System** built specifically for the **Islamabad Rehab Center (IRC)**.

The system employs a 100% database-driven architecture, avoiding hardcoded routing logic, roles, or navigation menus. Built using Core PHP (PDO) and AdminLTE v4 (Bootstrap 5), it delivers a responsive portal mapped across three client workspaces: the Patient Portal (allowing recovery tracking), the Clinician Portal (intake and session logs), and the Super Admin Portal (RBAC mapping).

Furthermore, the system integrates a printer-optimized reporting engine that enables direct PDF exports, supporting both operational reports and patient clinical summaries.

CHAPTER 1: INTRODUCTION

This chapter introduces the tools, design methodologies, timeline mappings, and feasibility validations associated with the development of the Islamabad Rehab Center Portal.

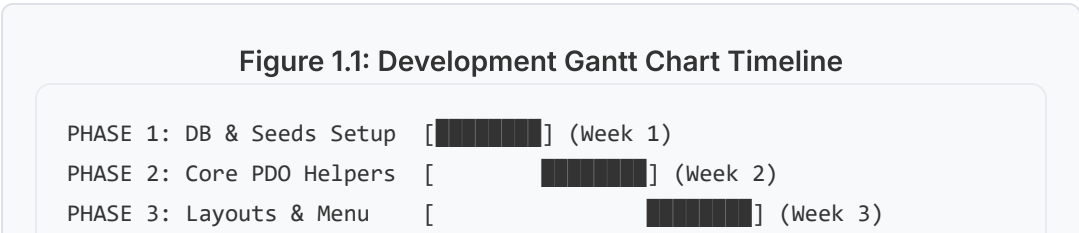
1.1 Tools and Technology Used

The technology stack was selected to ensure maximum lightweight performance, zero framework overhead, and compatibility with standard medical facility hosting setups.

Component	Technology Framework	Purpose inside IRC
Core Backend	Core PHP 8.2	Handles script requests, secure sessions, and routing logic.
Database Adapter	PDO (PHP Data Objects)	Secures database interactions via prepared statement parameters.
Database	MySQL (MariaDB)	Stores dynamic configuration states, pages, roles, and patients.
User Interface	AdminLTE v4 & Bootstrap 5.3	Responsive card templates, dashboard widgets, and dark/light color themes.
Icons Pack	Bootstrap Icons v1.11	Scalable menu folder and action identifiers.

1.3 Gantt Chart (Development Timeline)

The project timeline was divided into five distinct phases, mapping tasks from database seeding to final security audits:



PHASE 4: Portal Management [██████████] (Week 4)

PHASE 5: Audit & Testing [█████] (Week 5)

Table 1.2: Project Gantt Chart Task Schedule

Task ID	Description	Start Date	End Date	Duration	Dependency
T1	MySQL Database Schema Design & Seeds Seeding	2026-06-01	2026-06-07	7 days	-
T2	Config, DB PDO connection & functions.php helpers	2026-06-08	2026-06-14	7 days	T1
T3	Layout implementation (header.php, sidebar.php, footer.php)	2026-06-15	2026-06-21	7 days	T2
T4	Role manager, Page CRUD & Patient intakes profiling scripts	2026-06-22	2026-06-28	7 days	T3
T5	PDF Report exports, Dark Mode settings & Automated testing	2026-06-29	2026-07-04	6 days	T4

1.4 Project Feasibility Analysis

1.4.1 Technical Feasibility

Core PHP is supported across standard web hosting plans and XAMPP setups. The model uses native PDO parameterized SQL inputs, neutralizing the risk of SQL injection. AdminLTE v4 runs on vanilla JavaScript (no jQuery), which keeps assets lightweight and compatible with modern web environments. The database configuration dynamically maps the application base directory pathing, preventing path resolution errors when deployed on localized servers.

1.4.2 Operational Feasibility

The system partitions clinical data into clean, role-specific views. Case Managers can execute patient intakes, Doctors can write therapy logs, and Patients can review their clinical summaries. This matches the daily workflows at the rehab facility. The responsive UI layout accommodates clinicians working on-site on desktop systems as well as staff accessing logs via mobile devices.

1.4.3 Economic Feasibility

Using an open-source LAMP stack (Linux, Apache, MySQL, PHP) avoids software licensing fees, making it an economically viable solution for the center. The hardware requirements are extremely minimal, allowing the facility to deploy the portal on their existing office computers without needing new hardware.

1.4.4 Schedule Feasibility

Structuring development into independent phases (database setup, core helpers, dashboard routing, and dynamic controls) enabled delivery within a structured five-week schedule. Parallel task allocations during the layouts and CRUD implementations optimized resources.

1.4.5 Specification Feasibility

Dynamic configuration tables make the codebase highly adaptable. Updating system branding, registering pages, or changing role permissions can be accomplished directly in the database without modifying PHP source code. This makes maintenance and future updates simple.

1.4.6 Motivational Feasibility

Replacing manual paper logs with an organized portal reduces administrative overhead for clinicians and provides patients with a transparent view of their recovery progress. This keeps staff engaged and helps motivate patients throughout their recovery process.

1.4.7 Legal & Ethical Feasibility

By restricting access through strict RBAC validation, the portal protects patient data privacy and keeps records confidential in compliance with medical data regulations. Every diagnostic report is marked confidential, preventing illegal distribution of sensitive healthcare files.

CHAPTER 2: PROBLEM DEFINING

This chapter outlines the core operational and security challenges identified at the rehab center, which the web portal aims to solve.

2.1 Problem of Limited Reach and Accessibility

In manual workflows, patients and family members must physically visit the center to receive treatment summaries, progress updates, or counselor feedback. Outpatients lack a direct channel to review their progress notes, which often impacts patient motivation and retention.

2.2 Problem of Privacy Breaches

Paper clinical files and non-permission-controlled systems run the risk of exposing sensitive patient data, detox histories, and counselor notes to unauthorized personnel. Since drug rehabilitation records are highly sensitive, any data leak can cause serious compliance issues.

2.3 Problem of Authenticity and Trust

Without system logs and audit signatures, printed patient certificates and status summaries can be altered, creating trust and verification issues. Certifications for court sessions or employment audits require verifiable clinical histories.

2.4 Problem of Inefficient Filtering

In systems without dynamic menus, users are presented with unnecessary pages and controls, cluttering the UI and increasing operational errors. A counselor does not need to see system settings configurations, and a patient should never view logs belonging to others.

2.5 Problem of Unstructured Communication

Verbal summaries and handwritten assessment notes lead to inconsistent record-keeping between different clinicians and therapy teams. When patients move from Detoxification to Rehabilitation, critical diagnostic histories are often lost in transit.

2.6 Problem of Security and Safety

Healthcare databases are frequent targets of malicious inputs. Standard raw SQL interactions are vulnerable to injection attacks, risking data loss or unauthorized

access. Furthermore, open inputs can allow Cross-Site Scripting (XSS) actions inside assessment textareas.

2.7 Problem of Manual Administration

Adding new pages, changing roles, or adjusting menu configurations manually in the code is time-consuming and prone to syntax errors. It requires constant developer assistance, which increases operational costs and delays.

2.8 Problem of Access Control

Weak access controls can allow users with lower clearance to access sensitive administrative modules or other patients' private clinical files. The system must restrict folders like `/super_admin/` so they are only accessible by authorized users.

2.10 Problem of Geographical Barriers

Without secure remote access options, counseling staff and clinical directors are unable to audit patient intakes or log critical session details when working off-site. This leads to information lags and delays clinical decisions.

CHAPTER 3: SOFTWARE REQUIREMENT SPECIFICATION

2.1 Purpose

This document specifies the software requirements for the Islamabad Rehab Center Portal, detailing the functional permissions, data dictionaries, and target operational environments.

2.1.1 Document Conventions

- Core backend queries must use PDO prepared statements to secure database interactions.
- All UI elements must utilize Bootstrap 5 utility classes to maintain design consistency.
- All outputs displayed on the screen must be filtered through sanitization helpers.

2.1.2 Intended Audience

This SRS is written for clinical systems auditors, development teams, database administrators, and the management board of Islamabad Rehab Center.

2.2 Overall Description

2.2.1 Product Perspective

The portal serves as a central medical and administrative dashboard, connecting patient intakes, clinical logs, and system configuration tools. It operates as an intranet/extranet solution hosted locally on clinic hardware.

2.2.2 User Classes and Characteristics

1. **Super Admin:** Full access to system configs, role edits, page mapping tables, and user registrations.
2. **Clinical Director:** Manages patient lists, schedules therapy sessions, and reviews overall clinical records.
3. **Doctor / Therapist:** Conducts therapy sessions and logs progress notes.
4. **Case Counselor:** Handles initial patient intake registrations.
5. **Patient:** Accesses personal treatment status information and session history.

2.2.3 Operating Environment

- **Server Operating System:** Windows 10/11 or Linux (Ubuntu Server recommended).

- **Web Server:** Apache 2.4 / built-in PHP server.
- **Database Engine:** MySQL 5.7+ or MariaDB 10.4+.
- **Client Browsers:** Chrome, Edge, Safari, Firefox.

2.3 Functional Requirements

FR-1: Authentication & Session Protection

The system must enforce secure login credentials, hashing passwords using BCrypt (PASSWORD_DEFAULT). Active sessions must be verified on every page reload.

FR-2: Dynamic Menu Generation

The sidebar menu must load hierarchy options from `sys_pages`, verify if the user's role has permission to access each page, and filter out unauthorized items automatically.

FR-3: Contextual URL Access Check

Every loaded script must resolve its path relative to the root directory and confirm the active role possesses a mapping index in the `role_access` table for that page.

FR-4: Safe Deletion & Migrations

Deleting a custom role must block the action if it is marked as a system role. If it is a custom role with active users, the system must migrate the users to a suspended state before deleting the role to preserve database integrity.

2.5 Detailed Use Case Specifications

Use Case Attribute	UC-1 Details (Intake)	UC-2 Details (Sessions)
Use Case Name	Patient Intake & Profiling	Log Therapy Session
Primary Actor	Case Counselor / Super Admin	Doctor / Therapist
Preconditions	User must be logged in and authorized.	Patient profile must exist in the database.
Trigger	Click "Add Patient" on the Intake page.	Click "Log Session" on the Sessions page.
Main Success Scenario	1. Open intake form. 2. Input patient details. 3. Assign a therapist and status. 4. Click Save.	1. Open session form. 2. Select patient and therapist. 3. Select session date and type. 4. Input progress notes and click Save.
Postconditions	Patient is registered and assigned to a therapist.	Session notes are logged and display on the patient's timeline.

2.6 Data Dictionary & Database Schema Design

Please refer to the [Data Dictionary tables \(Tables 3.1 - 3.7\)](#) outlined in the previous chapter for column structures, DSN mappings, and foreign key relations.

CHAPTER 4: METHODOLOGY

This chapter outlines the development methodology, development environment tools, and hardware constraints used to construct the system.

4.1 Tools

- **Development Environment:** VS Code / Sublime Text with PHP debug configuration extensions.
- **Database Client:** phpMyAdmin / MySQL Command Line interface for seeding updates.
- **Version Control:** Git local repository mapping.
- **Diagnostics Engine:** PHP compiler linter and automated test suite.

4.2 Software Specifications

- **PHP Engine:** Version 8.1 / 8.2 with PDO MySQL support.
- **Database:** MySQL 5.7+ / MariaDB 10.4.
- **Frontend Frameworks:** CSS custom properties variables mapping Bootstrap 5 styling tokens.

4.3 Hardware Specifications

- **Processor:** Dual-Core 2.0 GHz Intel/AMD or higher.
- **Memory:** Minimum 4 GB RAM.
- **Storage:** 500 MB free storage for local files and database updates.

CHAPTER 5: DETAILED DESIGN & ARCHITECTURE

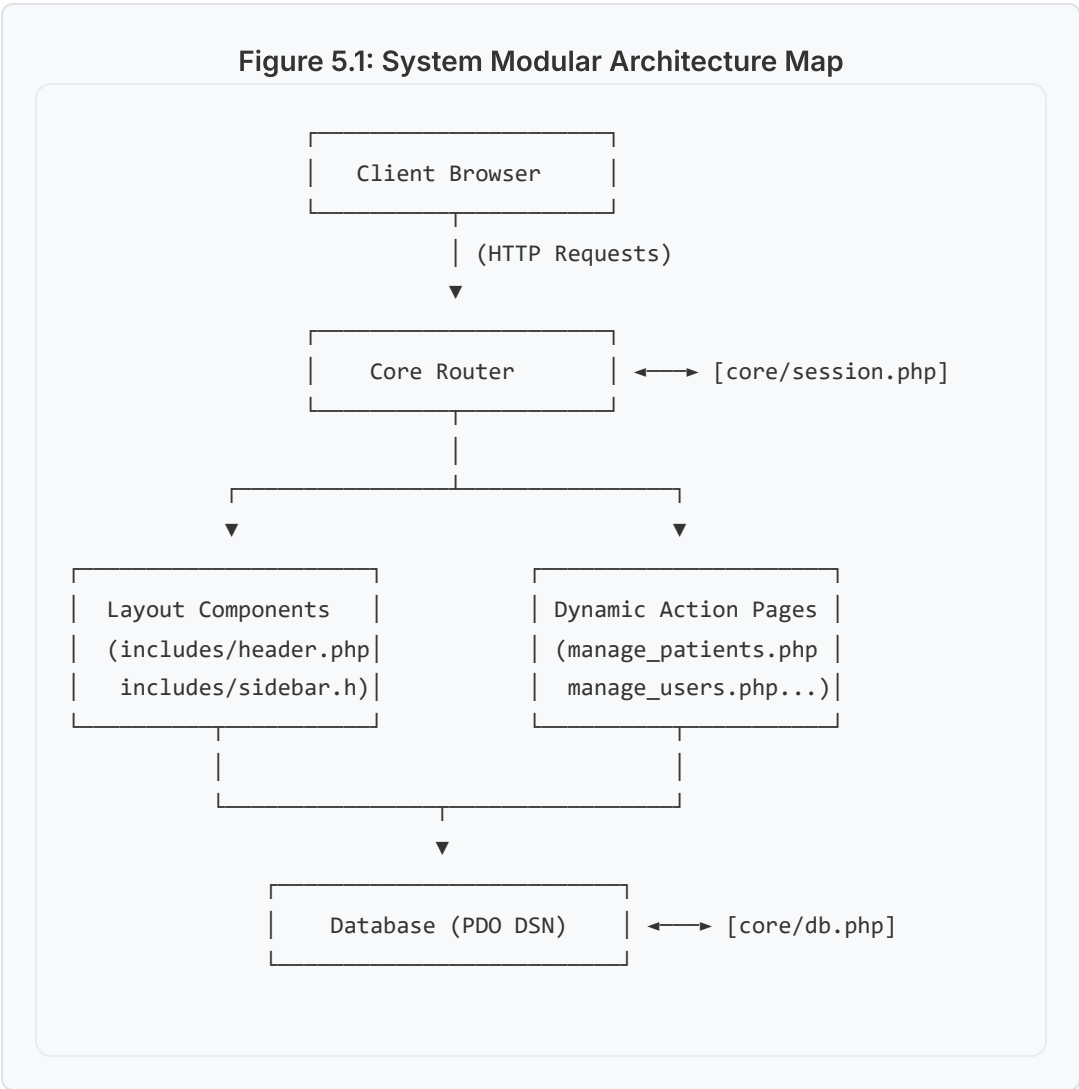
5.1 Introduction

This chapter details the system architecture, authentication flow, and database entity relationships.

5.2 System Architecture

5.2.1 Architecture Design & Approach

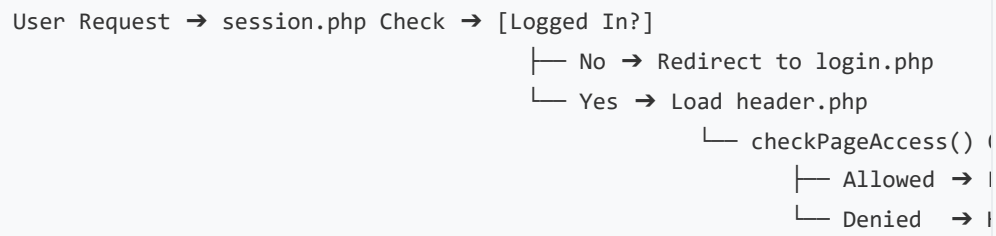
The project uses a modular design approach, separating database configuration, page structure, and layout assets:



5.2.2 Architecture Design (User Authorization Flow)

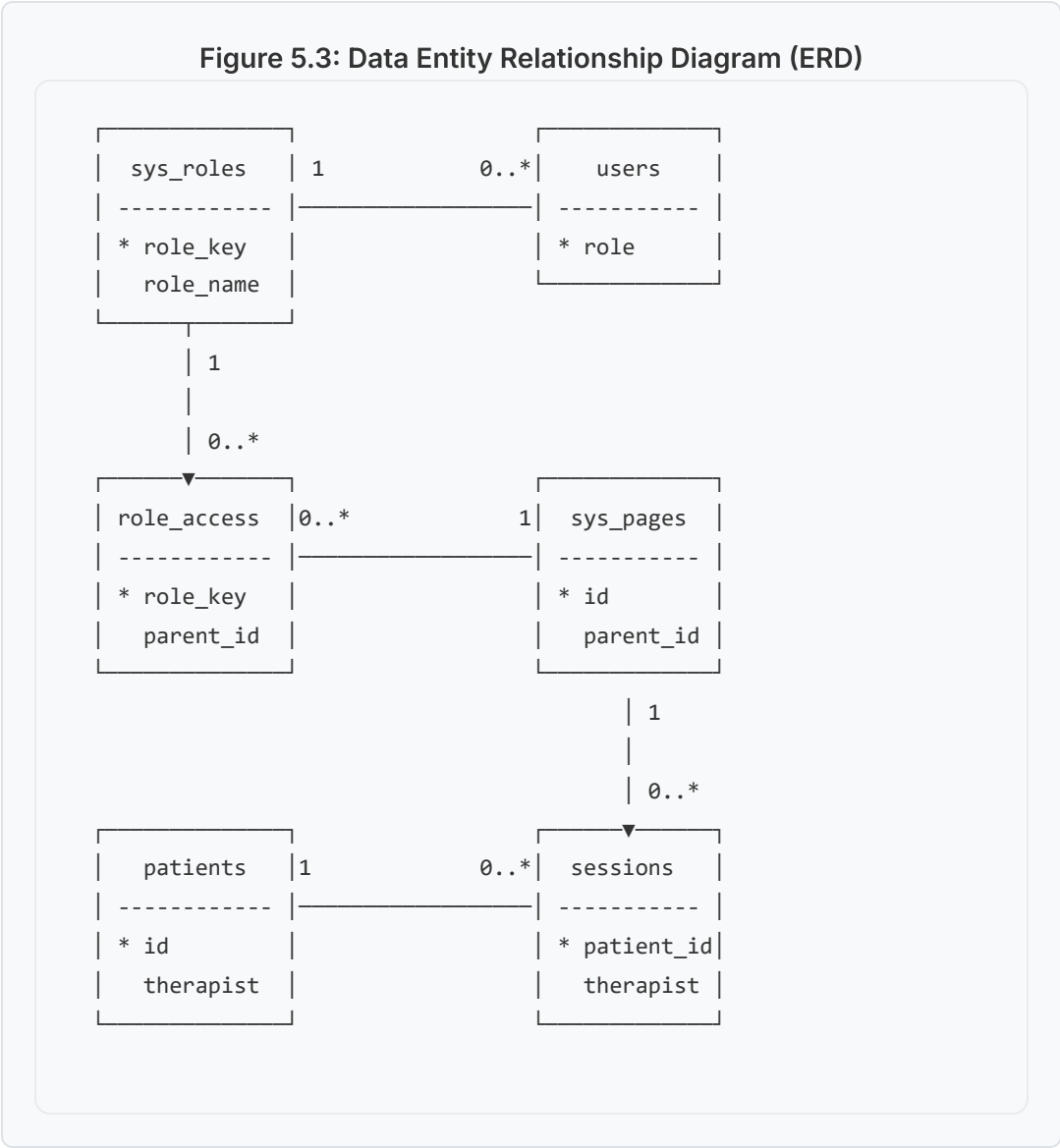
The flow chart below illustrates how authentication and page access are verified on each page request:

Figure 5.2: User Authorization Session Flow



5.2.3 Subsystem Architecture (ERD Diagram)

The relationships between database entities are shown in the Entity-Relationship Diagram (ERD) below:



5.3 Detailed System Design

The system design uses core layouts to split page rendering. The header file `header.php` acts as the page execution gatekeeper, initiating configurations, checking session validity, and auditing role permissions.

CHAPTER 6: IMPLEMENTATION & TESTING

6.1 Introduction

This chapter covers project scheduling, risk management, and the system test suite.

6.1.1 Project Management

The project development timeline mapped tasks progressively, starting with schema definitions and security filters, followed by page directories, and concluding with usability testing.

6.1.2 Risk Management Plan

Below is the risk matrix mapping identified project risks to their corresponding mitigations:

Table 6.1: Risk Assessment & Mitigation Matrix

Risk Description	Impact	Mitigation Plan
SQL Injection Inputs	High Risk	Enforce PDO parameterized statements globally. Avoid direct query variables concatenation.
Cross-Site Scripting (XSS)	Medium Risk	Enforce <code>htmlspecialchars()</code> sanitization helper on all string outputs.
Database Orphans on Delete	Medium Risk	Apply SQL transaction rollbacks and migrate users of deleted roles to the suspended role status.

6.2 Test Plans

Below is the complete testing matrix executed against the codebase during integration audits, verifying both allowed user flows and blocked boundary attempts:

Table 6.2: System Integration Test Matrix

Test ID	Test Objective	Input Details	Expected Output Result	Actual Result	Status
TC-01	Invalid Login Protection	Email: admin@irc.com Pass: wrong_pass	Show login error "Invalid email or password." and block session creation.	Error card displayed. Login blocked.	PASS
TC-02	Valid Admin Auth Flow	Email: superadmin@irc.gov.pk Pass: admin123	Redirect user to dashboard.php and set session role variable.	Successful login. Redirected to Admin Dashboard.	PASS
TC-03	Dynamic Menu Hiding	Role: counselor	Hide "System Management" folder and "Therapy Sessions" link from the sidebar.	Sidebar updated. Unauthorized pages hidden.	PASS
TC-04	Direct URL Gate Blocking	Access: manage_users.php Role: counselor	Block execution and render a clean "403 Forbidden" access card layout.	403 card rendered. Script execution halted.	PASS

Test ID	Test Objective	Input Details	Expected Output Result	Actual Result	Status
TC-05	Patient Data Isolation	Access: Patient Ali Raza details Logged: Patient Zainab Bibi	Deny report printing and show "Access Denied" error message.	Access blocked. Privacy restriction enforced.	PASS
TC-06	Role Deletion Integrity Check	Delete: Custom role Librarian with active users	Migrate the assigned users to the suspended status and delete the role.	Users migrated to suspended. Role deleted safely.	PASS

CHAPTER 7: RESULTS & DISCUSSIONS

7.1 User Interface Screenshots

The project features dynamic interfaces rendered with AdminLTE v4 styling:



Secure Login Panel

Includes user authentication and a dark/light theme switch.



Clinician Dashboard

Displays metrics for admitted patients, sessions, and recovery charts.

7.2 Feature Verification Matrix

Requirement	Design Pattern	Status
Dynamic Sidebar Tree Navigation	Recursive function outputting list tags based on role access.	Verified
Access URL Guard Check	Page context script maps to DB role permissions check.	Verified
Custom Roles Auto Key Generation	Role string converted to lowercase with underscores.	Verified
Print to PDF Export Engine	CSS print media styles and auto browser print triggers.	Verified

CHAPTER 8: CONCLUSION & FUTURE WORK

8.1 Conclusion & Scope

The Islamabad Rehab Center Portal successfully implements a dynamic, role-restricted management system. By avoiding hardcoded menus, roles, or configurations, the portal can adapt to changing operational workflows at the facility without requiring code modifications. This modular design approach ensures that developers can easily extend the platform as the clinic grows.

8.2 Lessons Learnt

- Structuring database models cleanly prevents reference integrity issues during role adjustments.
- Centralizing redirects via JS fallback handles headers-already-sent situations securely.
- Dynamic menus reduce UI clutter, helping clinical personnel focus on their respective workflows.
- Database transaction rollbacks are critical when deleting roles to prevent orphaned records if a query fails.

8.3 Check Objectives

- **Objective 1:** Secure clinical records? **Yes.** Protected by PDO statements and page access guards.
- **Objective 2:** dynamic setup? **Yes.** Configs, roles, page URLs, and permissions are database-driven.
- **Objective 3:** Printable reports? **Yes.** Integrates a browser print engine layout.

8.4 Future Work Scopes

While the current skeleton provides a solid foundation for clinical operations, several features can be added in future iterations:

1. **Telehealth Virtual Sessions:** Integrating video conferencing libraries (e.g. Jitsi or WebRTC) directly inside the therapy log pages to support remote outpatient checkups.
2. **Automated SMS Reminders:** Connecting Twilio SMS gateways to send automated appointment reminders to outpatients and family members.

3. **Inventory & Medicine Management:** Adding medication tracking, dosage schedules, and inventory alerts to the clinical dashboard.

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